

A Project Report on
Automated Mobile Robotic Arm for
Cold Storage Operations

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CERTIFICATE

This is to certify that project titled “**Automated Mobile Robotic Arm for Cold Storage Operations**” submitted by **Bhosale Subodh Shirish, Patil Girish Ashok, Jadhav Utkarsh Santosh, Pore Omkar Ashok** in partial fulfillment for the award of Bachelor of Mechanical Engineering of Savitribai Phule Pune University (S.P.P.U) at Vidya Pratishthan's Kamalnayan Bajaj Institute of Engineering and Technology, Baramati, is a record of original research work carried out under my supervision. To the best of our knowledge, the matter embodied in the thesis has not been submitted to any University/Institute for the award of any Degree.

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Abstract

This project focuses on the development of an Automated Mobile Robotic Arm (AMRA) designed specifically for cold storage operations. The primary goal of this system is to enhance the efficiency of item handling and transportation in environments with extreme temperatures, thereby reducing the need for manual labor and mitigating health risks to workers. The robotic arm is equipped with high-torque MG996R servo motors and a temperature-resistant battery (Samsung Li-Ion 7.4V 2600mAh), making it suitable for use in cold storage conditions. The system also includes a servo controller that enables precise and smooth control of the arm's movement through Bluetooth connectivity.

The robotic arm operates on a mobile platform, allowing it to navigate storage aisles and adapt to various layouts, making it highly flexible and efficient in real-time operations. It is designed to perform tasks such as retrieving, placing, and transporting items within a cold storage facility, while ensuring that the items are handled with the necessary precision and care required for temperature-sensitive goods.

The project also addresses key challenges such as the impact of low temperatures on system performance and longevity. To mitigate these issues, the robotic arm uses special materials, lubricants, and AI-based path planning algorithms to optimize movement and reduce energy consumption, ensuring stable operation in sub-zero environments.

This document outlines the methodology, hardware components, and software requirements for the robotic arm, including the design and testing process. It also details the future steps of system integration, cold storage environment simulation, and performance testing to ensure the robotic arm's robustness and reliability.

In conclusion, this project provides a comprehensive approach to developing a mobile robotic arm tailored for cold storage operations, with the potential to significantly improve operational efficiency and worker safety in temperature-controlled environments.

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Nomenclatures

AMRA	Automated Mobile Robotic Arm
MG996R	Servo Motor Model MG996R
SG90	Micro Servo Model SG90
Li-Ion	Lithium-Ion
mAh	Milliampere-Hour
PWM	Pulse Width Modulation
VDC	Volts Direct Current
IoT	Internet of Things
VDC	Voltage Direct Current
GPRS	General Packet Radio Service
RTOS	Real-Time Operating System
BMS	Battery Management System
LED	Light Emitting Diode
GSM	Global System for Mobile Communication
IR	Infrared
GPS	Global Positioning System
LIDAR	Light Detection and Ranging
FMEA	Failure Modes and Effects Analysis
PLC	Programmable Logic Controller
IMU	Inertial Measurement Unit
PID	Proportional Integral Derivative (Control
CAD	Algorithm)
AI	Computer-Aided Design
	Artificial Intelligence

- **Problem Statement:**

In cold storage environments, handling and transporting goods is a challenging task due to extreme temperatures. The manual movement of temperature-sensitive items is labor-intensive and can lead to increased risks to worker health and safety, as well as reduced operational efficiency. Existing automated systems are often limited to stationary functions or have low mobility, making them less adaptable to the dynamic needs of cold storage facilities. This project addresses these challenges by developing an automated mobile robotic arm capable of operating in cold storage environments. The system aims to improve productivity, reduce manual labor, and enhance safety while maintaining the precision and durability required for such harsh conditions.

- **Aim:**

The primary aim of this project is to develop a fully functional Automated Mobile Robotic Arm (AMRA) for efficient, precise, and autonomous operation within cold storage environments. The AMRA aims to improve overall productivity, reduce human exposure to extreme temperatures, enhance safety in cold storage operations, and optimize the movement and handling of temperature-sensitive products.

By integrating mobility with high-precision robotics, the system will ensure reliable and accurate operation in sub-zero conditions while maintaining efficiency. The project also aims to address challenges such as power consumption, motor durability, and the robustness of materials, ensuring the robotic arm remains operational in harsh cold storage conditions.

- **Objectives:**

1. Design and Development: To design a robotic arm integrated with wheels that can navigate cold storage environments and perform tasks such as retrieving, placing, and transporting items.

2. Cold Storage Adaptation: To select and use materials and components (such as motors, sensors, and batteries) that are durable and effective under low-temperature conditions.
3. Increase Efficiency: To improve the speed and precision of item handling in cold storage environments, reducing manual labor and increasing throughput.
4. Safety Enhancement: To ensure the robotic arm reduces the risk of injury to workers by automating physically demanding tasks in hazardous temperature conditions.
5. System Testing and Optimization: To test the robotic arm in simulated cold storage environments and fine-tune its movement, control systems, and navigation to ensure optimal performance.
6. Integration with Advanced Technologies: To explore the integration of AI-based path planning, machine vision, and automated navigation to further enhance the system's performance and efficiency.

By achieving these objectives, the project aims to create a versatile and reliable solution for automation in cold storage logistics, enhancing both operational efficiency and worker safety.

1. INTRODUCTION

Cold storage facilities play a crucial role in industries like food processing, pharmaceuticals, and biotechnology, where temperature-sensitive goods need to be stored and transported under controlled conditions. However, the extreme low temperatures in these environments create significant challenges for traditional manual handling and existing automation systems. Workers face health risks, and productivity is often slow and labor-intensive.

To address these challenges, automation in cold storage is increasingly becoming a necessity. Automated systems, such as robotic arms and mobile platforms, can enhance efficiency, reduce human exposure to cold, and improve safety. However, most existing systems are not designed for the harsh conditions of cold storage, limiting their effectiveness.

This project focuses on the development of an “Automated Mobile Robotic Arm (AMRA)” for cold storage environments. The goal is to design a robotic arm capable of autonomously retrieving, placing, and transporting items with high precision in sub-zero temperatures. The robotic arm will be mounted on a mobile platform, allowing it to navigate aisles, avoid obstacles, and adapt to different layouts within the cold storage facility.

The aim is to improve operational efficiency, reduce labor costs, and enhance worker safety by minimizing human intervention. This system will also ensure the careful handling of temperature-sensitive goods, thus improving the overall productivity and safety of cold storage operations. The following sections will detail the design, development, and testing of the AMRA, focusing on its hardware, software, and integration into a cold storage environment.

1.1 Need for Automated Systems in Cold Storage Operations

1.1.1 Challenges in Manual Cold Storage Operations

1. Extreme Working Conditions: Cold storage environments typically operate at sub-zero temperatures, making them challenging for human workers. Prolonged exposure to freezing conditions can lead to fatigue, reduced efficiency, and health risks, such as frostbite or hypothermia.

2. **Productivity Limitations:** Human workers in cold storage are often limited in their ability to perform repetitive tasks efficiently over extended periods. Manual operations also face challenges in meeting the increasing demand for faster order processing in logistics and supply chains.
3. **High Error Rates:** Manual handling of inventory in cold storage can result in errors such as incorrect item picking, misplacement, or damage to goods. These mistakes can lead to financial losses and delays in the supply chain.
4. **Labor Shortages:** Cold storage facilities often struggle to find and retain workers due to the demanding nature of the job, leading to increased operational costs and disruptions in warehouse activities.

1.1.2 Benefits of Automation in Cold Storage

1. **Enhanced Safety:** Automated systems reduce the need for human workers to operate in freezing conditions, minimizing health risks and improving overall workplace safety. Robots and automated guided vehicles (AGVs) can handle tasks in hazardous environments, ensuring uninterrupted operations.
2. **Increased Efficiency:** Automation enables faster and more accurate inventory management, retrieval, and placement. Robots can perform repetitive tasks continuously without breaks, significantly increasing productivity compared to manual operations.
3. **Improved Accuracy:** Automated systems equipped with AI and machine vision can precisely identify, track, and handle items, reducing errors and ensuring that inventory is accurately managed.
4. **Cost Savings:** While initial investments in automation are high, long-term savings can be achieved through reduced labor costs, lower error rates, and decreased product wastage due to improved handling and storage practices.
5. **Scalability and Adaptability:** Automated systems can easily scale to meet growing demands. They are also adaptable to changing warehouse layouts and storage needs, providing flexibility for future expansion.

1.2 History of Robotics in Cold Storage

1.2.1 Early Developments

The history of automation in cold storage can be traced back to the early 20th century when the first rudimentary systems were introduced to handle temperature-sensitive products. These initial systems were primarily focused on refrigeration and basic mechanical automation, such as conveyor belts and simple automated storage racks. These early machines were designed to reduce manual labor in cold storage facilities, but their capabilities were limited by the technology of the time, especially in terms of mobility and precision.



Fig.1.1. History of Robotics in Cold Storage

1.2.2 The Rise of Online Systems

In the latter half of the 20th century, as industries grew, the demand for more sophisticated solutions increased. The introduction of online systems, such as Automated Storage and Retrieval Systems (ASRS), allowed for more efficient storage and retrieval of goods. These systems typically operated within fixed aisles and required little human intervention for retrieving goods from cold storage. However, they were still limited in scope and flexibility, as they could only work in a predetermined space and were not designed for cold storage environments that required specialized systems to handle low temperatures.



Fig.1.2. The Rise of Online Systems

1.2.3 The Emergence of Offline Systems

By the 1990s, the development of offline automated systems began to gain traction. These systems featured more flexible configurations compared to online systems and were capable of being deployed in a wider range of environments, including cold storage. Automated guided vehicles (AGVs) and robotic arms began to emerge, although their adoption in cold storage environments was limited due to challenges in adapting them to work effectively in freezing conditions. Despite these challenges, offline systems offered greater mobility, and their use in logistics and warehouses started to rise.

1.2.4 The Rise of Robotics and Autonomous Systems

The early 2000s saw a significant shift with the rise of advanced robotics and autonomous systems. Robotic arms, mobile robots, and drones began to take center stage in automating not just cold storage, but logistics and warehousing in general. These systems were designed to be flexible, mobile, and capable of operating in a variety of environments, including sub-zero temperatures. They utilized advanced sensors, AI, and machine learning algorithms for task management, real-time decision-making, and autonomous navigation. The rise

of autonomous mobile robots (AMRs) allowed for more complex operations such as inventory management, item picking, and delivery without human intervention, even in the cold storage sector.

1.2.5 Current Trends and Future Directions

Today, cold storage facilities are increasingly adopting robotic and automated systems, with trends focusing on greater efficiency, energy savings, and reducing human exposure to harsh conditions. Modern systems are integrating artificial intelligence (AI) and machine vision, enabling robots to not only navigate autonomously but also to identify products, scan barcodes, and handle goods more intelligently. The development of temperature-resistant materials and specialized components for cold environments has further pushed the capabilities of robotic systems in these areas. Looking ahead, the future of automated robotic systems in cold storage lies in the integration of Internet of Things (IoT) technology, predictive maintenance, and even more sophisticated AI-driven decision-making, paving the way for fully autonomous cold storage facilities. The ongoing trend toward flexibility, scalability, and cost-effectiveness will likely drive further innovation in the field, making cold storage more efficient and safer for workers.

1.3 Challenges in Cold Storage Environments

1.3.1 Harsh Environmental Conditions

1. Sub-Zero Temperatures: Cold storage facilities often operate at temperatures as low as -20°C to -30°C , posing significant challenges for both humans and machinery. Prolonged exposure to such conditions can lead to equipment malfunctions and health issues for workers.
2. Condensation and Frost: Rapid temperature changes can cause condensation and frost buildup, which can affect product quality, reduce visibility, and compromise the functionality of equipment like sensors and conveyor belts.
3. Energy Consumption: Maintaining a consistent low temperature requires substantial energy input, leading to high operational costs and challenges in energy-efficient management.

1.3.2 Logistics and Inventory Management

1. Perishable Goods: Cold storage primarily handles perishable items such as food, pharmaceuticals, and chemicals, requiring precise temperature and humidity controls to ensure product integrity.
2. Space Optimization: Limited storage capacity in cold storage facilities demands efficient use of space while maintaining accessibility for frequent retrieval and placement of goods.
3. Complex Handling Requirements: Different products may have specific storage and handling requirements, increasing the complexity of warehouse operations

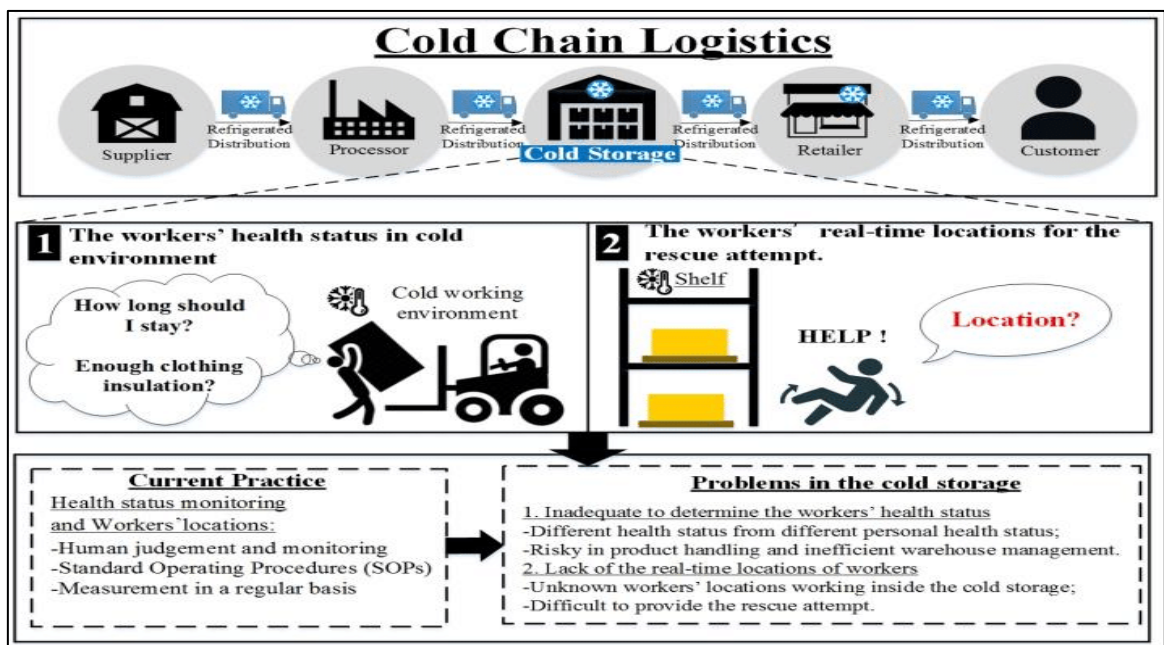


Fig.1.3. Logistics and Inventory Management

1.3.3 Workforce-Related Issues

1. Employee Retention: The physically demanding nature of work in cold storage leads to high turnover rates, making it difficult to maintain a stable workforce.
2. Limited Working Hours: Human workers can only operate in cold environments for short durations, necessitating frequent breaks and reducing overall productivity.

3. Skill Gaps: Handling sophisticated cold storage systems often requires technical expertise, which may not be readily available in the workforce.

1.4 Significance of Automation in Cold Storage

1.4.1 Addressing Human Limitations

1. Health and Safety: Automation eliminates the need for human workers to operate in freezing environments, reducing health risks and ensuring continuous operations.
2. Enhanced Productivity: Robots and automated systems can work 24/7 without breaks, significantly improving throughput and meeting the growing demands of industries like food and pharmaceuticals.

1.4.2 Efficiency and Accuracy

1. Optimized Inventory Management: Automated systems equipped with AI and machine vision ensure accurate inventory tracking, retrieval, and placement, reducing errors and enhancing operational efficiency.
2. Space Utilization: Automation enables the use of advanced storage systems, such as high-density racks and automated retrieval systems, optimizing space in cold storage facilities.
3. Real-Time Monitoring: Sensors and IoT-enabled devices provide real-time data on temperature, humidity, and inventory status, ensuring optimal conditions and proactive issue resolution.

1.4.3 Cost and Sustainability Benefits

1. Energy Efficiency: Automated systems can optimize energy use by precisely regulating operations, reducing unnecessary energy consumption.
2. Reduced Labor Costs: While initial investments in automation may be high, they lead to long-term savings by minimizing labor costs and improving operational efficiency.
3. Sustainability Goals: Automation facilitates better energy management and reduces product waste, contributing to sustainability efforts and regulatory compliance.

2. LITERATURE REVIEW

The literature review examines the application of robotics in cold storage environments, emphasizing the advancements and challenges in using robotic systems, particularly robotic arms and automated solutions, in handling temperature-sensitive goods.

2.1 Robotics in Cold Storage Environments

Challenge: Robotics in cold storage must handle low temperatures and humidity, affecting performance.

Integration: Automated Guided Vehicles (AGVs) with robotic arms have been explored to handle cold storage tasks, adapting to the environment through robust sensors and thermal protection.

Application: Use of AGVs aids in automating repetitive tasks, reducing manual labor, and improving safety by working in harsh conditions.

2.2 Previous Work on Robotic Arms in Logistics

Key Developments: Significant progress in mobile platforms with robotic arms, which allow for pick-and-place operations, object sorting, and handling.

Technological Advancements: Adoption of servo motors and advanced controllers (like Arduino) to increase precision and accuracy in logistics operations.

Applications: Warehouse and manufacturing automation, with a focus on object manipulation and packaging.

2.3 Automation in Temperature-Sensitive Goods Handling

Focus: Handling of sensitive goods using automated systems requires precision and reliability.

Solution: Integration of vision systems and sensors for monitoring environmental conditions to ensure safe handling of perishable items.

Trend: Increased use of mobile robotic platforms that adapt to varying environments to handle fragile and temperature-sensitive products.

2.4 Gaps in Existing Robotic Systems for Cold Storage

Durability Issues: Challenges in sensor calibration and reliability in extreme temperatures, leading to potential inaccuracies.

Design Gaps: Existing systems lack sufficient energy efficiency and thermal management for cold storage operations, requiring advancements in robotic arm material and structural robustness.

Research Need: Focus on creating systems that maintain precision in low-temperature conditions while enhancing energy efficiency and minimizing maintenance.

3. System Design and Methodology

3.1 Design of Robotic Arm

- Objective: Design a robotic arm capable of operating in low-temperature environments with precision and reliability.
- Design Approach:
 - CAD Modeling: Utilize software like SolidWorks or AutoCAD to design the arm, focusing on joint flexibility, strength, and minimal heat loss.
 - Material Selection: Use materials such as aluminum alloys or polymers with low thermal conductivity to prevent brittleness in cold environments.
 - Load Handling: Design to support a payload capacity defined by torque calculations, ensuring stable and precise movement.

3.2 Motor and Servo Selection for Cold Storage

- Selection Criteria:
 - Temperature Range: Components must function reliably in a range of -20°C to +60°C.
 - Torque and Precision: Ensure sufficient torque for lifting and precise positioning of items.
- Chosen Components:
 - MG996R Servo Motor: Torque of 11 kg/cm; operating temperature range of -10°C to +60°C.
 - Micro Servo SG90: For lightweight precision tasks, operating at standard low temperatures.
- Placement and Configuration:
 - Strategic placement of motors to balance weight and optimize energy consumption.

3.3 Battery and Power Management System

- Battery Selection:
 - Samsung Cell Premium Li-Ion 7.4V 2600mAh: Selected for its long-lasting performance in temperatures as low as -20°C.
- Power Management:
 - Include a Battery Management System (BMS) to monitor charge levels and prevent over-discharge.

Optimize power delivery for simultaneous motor operation without overheating.

- Backup Plan:

Consider an auxiliary power source for uninterrupted operation in case of battery failure.

3.4 Mobility and Navigation System

- Mobility Design:

Wheel System: Use high-grip wheels with insulation to avoid freezing or reduced traction on cold surfaces.

Chassis Design: A lightweight yet durable frame to house the robotic arm and other components.

- Navigation:

Integration of ultrasonic sensors and LiDAR for obstacle detection and path planning.

Use a Raspberry Pi or Arduino-based microcontroller for real-time navigation and decision-making.

- Cold Storage Considerations:

Insulated motor housings to prevent freezing of mobile components.

3.5 Control System and Software Integration

- Hardware:

Use a servo controller compatible with PC or Bluetooth for intuitive operation.

Employ sensors (e.g., temperature, position encoders) for precise feedback control.

- Software:

Programmed in Python or C++ for efficient motor control, navigation, and task automation.

Real-time feedback loop for monitoring and adjusting operations during cold storage tasks.

- Interface:

Develop a user-friendly GUI to allow operators to set tasks, monitor status, and troubleshoot remotely.

3.6 Testing and Evaluation Protocol

- Simulated Environment:
Use a cold chamber to replicate real-world cold storage conditions, monitoring the system's performance.
- Key Testing Parameters:
Torque and load handling of the robotic arm.
Power consumption and battery life under continuous operation.
Stability and mobility on cold, slippery surfaces.
- Evaluation Metrics:
Efficiency of item handling and placement.
Response time to control commands.
Reliability and robustness after prolonged exposure to low temperatures.
- Iteration:
Perform iterative testing to refine mechanical design, software control, and overall system integration.

4. HARDWARE COMPONENTS

4.1. MG996R Servo Motors

- Working: The MG996R motor rotates its shaft based on received electrical signals. It uses feedback from an internal potentiometer to adjust its position and provide accurate control.
- Why It Is Used: The MG996R is selected for its high torque (11 kg/cm) and temperature resistance (-10°C to +60°C), making it ideal for the cold storage environment. Its precise movement capability ensures the arm can handle moderate loads and perform delicate tasks accurately in low temperatures.



Fig.4.1. MG996R Servo Motors

Operating voltage	4.8 V - 7.2 V
Running Current	500 mA – 900 mA (6V)
Stall Current	2.5 A (6V)
Temperature range	0 °C – 55 °C
Control System	Analog
Operating Angle	180degree
Required Pulse	900us-2100us
Direction	CCW

Table.1 specification of MG996R Servo Motor

4.2. SG90 Micro Servo Motor

- Working: This smaller servo motor operates similarly to the MG996R but with lower torque. It rotates to a precise angle when signaled by the controller.
- Why It Is Used: The SG90 is used for lighter tasks that require less force, such as handling smaller items. Its compact size and cold resistance (-10°C to $+60^{\circ}\text{C}$) make it perfect for tasks in tight spaces within cold storage areas.



Fig.4.2. SG90 Micro Servo Motor

4.3. Servo Controller

- Working: The servo controller communicates with the servo motors to control their movement. It interprets commands from a microcontroller or a PC and adjusts the motors accordingly. It typically uses Bluetooth for wireless communication.



Fig.4.3. Servo Controller

- Why It Is Used: This controller ensures precise and real-time motor control, allowing for smoother movement and easier programming. Bluetooth functionality provides convenient wireless control, essential for managing the robotic arm in a dynamic cold storage environment.

4.4. Samsung Cell Premium Li-Ion 7.4V 2600mAh Battery

- Working: The battery stores electrical energy and powers the robotic arm through a chemical process inside. It provides a stable voltage of 7.4V with a capacity of 2600mAh.



Fig.4.4. Samsung Cell Premium Li-Ion

- Why It Is Used: This Li-Ion battery is selected for its high energy capacity, ensuring longer operational times in cold environments. It also operates well in temperatures ranging from -20°C to $+60^{\circ}\text{C}$, making it suitable for cold storage operations without performance degradation.

4.5. Chassis and Wheels

- Working: The chassis serves as the structure of the robotic arm, supporting motors, sensors, and electronics. The wheels are powered by motors and enable the robotic arm to move around the cold storage area.
- Why It Is Used: The mobility provided by the wheels allows the arm to transport items efficiently. The chassis supports the entire robotic system,

ensuring stability and durability in cold conditions. The wheels are designed to perform well on icy or slippery surfaces commonly found in cold storage environments.



Fig.4.5. Chassis and Wheels

4.6. Sensors (Ultrasonic, Temperature, and Load Sensors)

- Working: Ultrasonic sensors detect obstacles by emitting sound waves and measuring the time taken for them to reflect back. Temperature sensors monitor the environmental temperature, while load sensors measure the weight of objects the robotic arm is carrying.
- Why It Is Used: Sensors enable the arm to avoid obstacles, monitor the cold storage environment, and ensure safe handling of items. The temperature sensors protect the system from extreme cold, and load sensors prevent overloading the arm, ensuring safe operations.

4.7. Control Unit (Microcontroller or Single-Board Computer)

- Working: The control unit processes inputs from sensors and issues commands to the servo motors, controlling the robotic arm's actions. It coordinates all components and performs tasks like navigation and item handling.

- Why It Is Used: The control unit acts as the brain of the robotic system, managing data and coordinating movements. It allows the robotic arm to operate autonomously or under remote control, ensuring smooth, real-time operations in the cold storage setting.

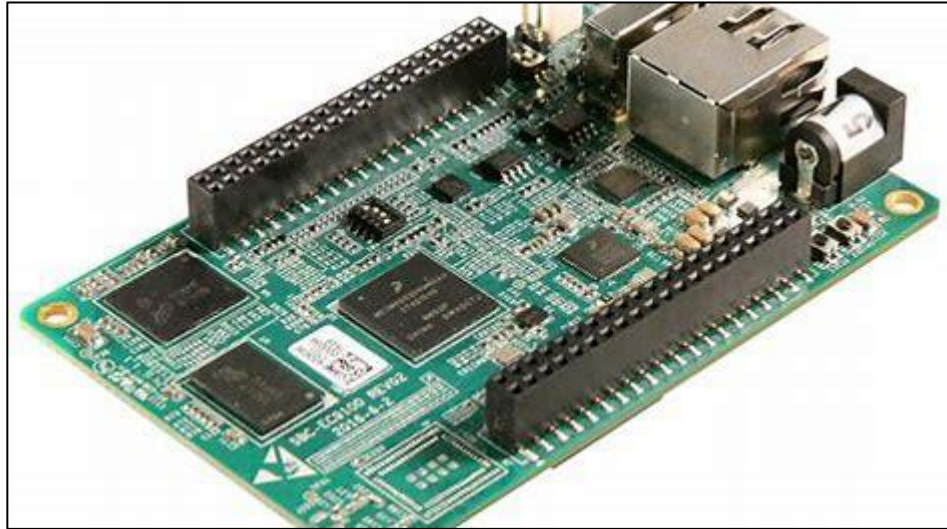


Fig.4.7. Control Unit (Microcontroller or Single-Board Computer)

4.8. Camera

- Working:

The camera captures visual data from the cold storage environment, such as the location and orientation of items to be handled by the robotic arm. This data is processed to identify and track objects, enabling precise movements and interactions. Advanced cameras may use techniques like object recognition, barcode reading, or depth sensing to enhance accuracy.
- Why It Is Used:

The camera functions as the "eyes" of the robotic system, providing critical visual input for decision-making. It enables the robotic arm to locate, identify, and manipulate items with precision, even in challenging cold storage conditions where human handling may be inefficient. This vision capability ensures the system operates efficiently and reliably while minimizing errors in item handling.

Component	Purpose
n.3 955 or 946 Servo Motors	Control the movement of different joints in the robotic arm with precision and torque.
n.1 SG90 Servo	Used for lightweight movement, often at the end effector (claw) for smaller tasks.
n.1 M6 Self-Locking Nut	Secures larger screws in place, preventing loosening during arm movement.
n.1 M6x25 Screw	Provides structural support for the arm's mechanical assembly.
n.2 M3 Self-Locking Nuts	Used to lock M3 screws securely in place, ensuring stable connections in the arm.
n.2 M3x20 Screws	Secures smaller components of the arm, often used in joint assemblies.
n.1 M3x10 Hex Recessed Screw	Holds smaller parts together while fitting into a hexagonal recess for added stability.
n.9 M4 Self-Locking Nuts	Prevents loosening of M4 screws during operation, ensuring durability of the structure.
n.1 M4x40 Screw	Structural support for larger segments of the arm.
n.1 M4x30 Screw	Connects parts of the arm that require slightly less length than the M4x40.
n.5 M4x20 Screws	Secures smaller joints or parts that need flexibility in movement.
n.1 M4x60mm Threaded Rod	Used to provide a pivot or rotational axis for the robotic arm's movement.
n.1 M4x32mm Threaded Rod	Provides a shorter rotational axis or structural support.
n.25 dia 6 mm Ball Spheres	Likely used in a bearing system to enable smooth rotation or movement.
n.1 606zz Bearing	Reduces friction in the rotating parts of the robotic arm, ensuring smoother motion.
Some M4 Washers	Distributes the load of screws and nuts, preventing damage to surfaces and ensuring secure fastening.

Table:2. Component and purpose

Name	Quantity
Control Board	1
Servo Motor MG 996r	3
micro Servo Motor	1
Stepper	2
Stepper Driver	4
battery	1
Filament	1
M6 Self Locking Nut	1
M6×25 Screw	1
M3 self-locking nut	2
M3×20 Screws	2
M3×10 Hex Recessed Head Screw	1
M4 Self-locking Nut	9
M4×40 Screw	1
M4×30 Screw	1
M4×60 mm Threaded Rod	1
M4×32 mm Threaded Rod	1
606zz Bearing	1
M4 Washers	15
M4×20	5
Dia 6 mm Ball Spheres	25

Table:3. Name of component and Quantity

5. **SOFTWARE REQUIREMENT**

5.1. **Arduino IDE (Integrated Development Environment)**

- Purpose: The Arduino IDE will be used for programming the microcontroller that controls the robotic arm's movements and functionality.
- Details:
Programming Languages: C/C++
Functionality:
 - The IDE will be used to write and upload code to Arduino-based microcontrollers (e.g., Arduino Uno, Mega) that control the servo motors and other components.
 - It simplifies interfacing with hardware components like servo motors (MG996R, SG90) and motor controllers, ensuring precise movement control.
 - Allows for real-time monitoring and debugging using the serial monitor.
- Why It's Used:
It's user-friendly, widely supported, and essential for developing and testing the robotic arm's software.

5.2 **PyCharm (Python IDE)**

- Purpose: PyCharm will be used for writing Python scripts to handle the higher-level logic of the robotic arm, such as path planning, sensor data processing, and communication with other software components.
- Details:
Programming Language: Python
Functionality:
 - PyCharm provides an IDE for coding scripts that interface with ROS and manage control algorithms.
 - Python scripts will handle tasks like controlling the movement of the robotic arm, managing sensor data, and integrating with other robotic systems.
 - Provides debugging and version control tools, which streamline development.
- Why It's Used:
Python is ideal for robotics due to its simplicity and extensive libraries. PyCharm helps ensure efficient coding and debugging.

5.3 CAD Software (SolidWorks)

- Purpose: CAD software will be used for designing the robotic arm's structure and other mechanical components.
- Details:
Functionality:
 - CAD tools like SolidWorks allow for the design of the robotic arm's structure, joints, grippers, and mobile base (wheels or tracks).
 - These designs can be exported for simulation and physical manufacturing.
 - These tools help create detailed technical drawings that guide fabrication and assembly.



Fig. 5.3 CAD Software (SolidWorks)

- Why It's Used:
Essential for designing mechanical systems and generating accurate models for 3D printing or machining.

5.4 Tinkercad

- Purpose: Tinkercad is an easy-to-use, online 3D design and 3D printing app.
- Details:
Functionality:
 - Tinkercad will be used to create basic prototypes and 3D models for components of the robotic arm, such as the gripper or joints.

- It allows for quick iterations and design testing, which is crucial during the prototyping phase.
- It also supports exporting models for 3D printing or integration with other tools.

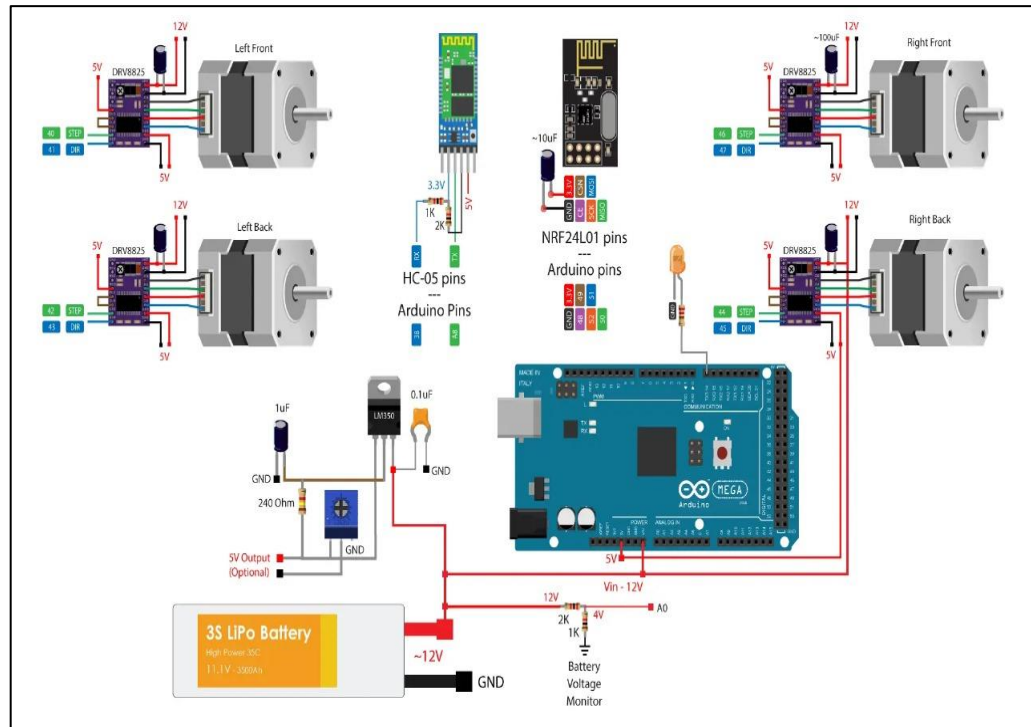


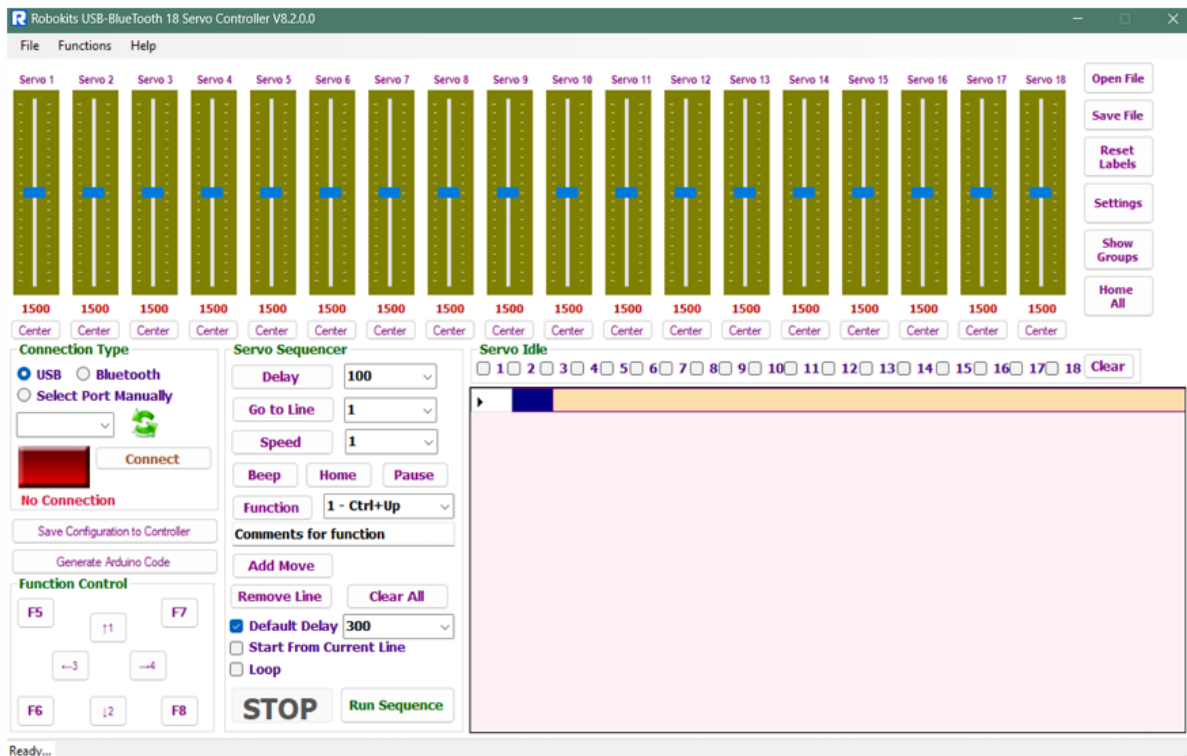
Fig.5.4 Tinkercad for Robotic Car

- Why It's Used:
Tinkercad is simple and free to use, making it a great tool for quick, low-complexity designs, especially for beginners or preliminary models.

5.5 RoboKits USB Bluetooth 18 Servo Controller Software

- Purpose: This software will be used to control the servo motors via Bluetooth communication.
- Details:
Functionality:
 - The Robo-Kits USB Bluetooth 18 Servo Controller software allows you to connect and control up to 18 servos through Bluetooth.

- It enables you to send commands from a PC or mobile device to the servo controller, which then translates them into motor actions for the robotic arm.



5.5 RoboKits USB Bluetooth 18 Servo Controller Software

- The software allows real-time control and adjustment of the arm's position and movements.
- Why It's Used:
It simplifies the control of multiple servo motors, making it easier to operate the robotic arm, particularly in a wireless configuration via Bluetooth.

6. Challenges and Solutions

6.1 Environmental Challenges in Cold Storage

Cold storage environments present significant challenges due to extreme low temperatures, which can affect the performance of electronic components, motors, and batteries. Standard robotic systems may not function optimally in such conditions, leading to reduced efficiency or failure of critical parts.

- Solution: The team selected MG996R servo motors and a Samsung Cell Premium Li-Ion battery, both of which are rated for cold environments. The motors can operate effectively between -10°C to $+60^{\circ}\text{C}$, and the battery performs well in temperatures as low as -20°C . These components were chosen after evaluating several options for their durability and performance in sub-zero conditions.

6.2 Overcoming Precision and Movement Issues

In cold storage, the robotic arm needs to handle items with high precision while being mobile, which can be a challenge in a harsh and slippery environment. Standard motors and control systems often struggle to provide the necessary torque and responsiveness in low temperatures, affecting the accuracy of movements.

- Solution: The MG996R servo motors provide a maximum torque of 11 kg/cm, ensuring that the arm can carry moderate loads with high precision even in colder environments. Additionally, a Bluetooth-controlled servo controller was integrated, allowing for real-time adjustments and smoother, more accurate control of the robotic arm's movements.

6.3 Power Consumption and Battery Life Optimization

Operating a robotic arm in a cold storage environment demands high power efficiency. The cold temperatures can lead to faster discharge rates for batteries, requiring careful selection and optimization of power sources to ensure prolonged operation.

- Solution: The Samsung Cell Premium Li-Ion 7.4V 2600mAh battery was selected due to its ability to sustain operations in temperatures as low as -20°C . Battery life is also optimized by employing low-power modes for the servo motors and using energy-efficient control algorithms, ensuring the system operates for longer periods without frequent recharges.

6.4 Software and Control Optimization

Programming the robotic arm for precise control in a cold storage environment can be challenging due to the complexity of managing motor positions and accounting for environmental variables like temperature-induced delays or performance degradation.

- Solution: The team utilized a servo controller that allows for easier control of the motors through a PC or laptop using Bluetooth technology. This setup enables real-time adjustments to the arm's movements, making it more adaptable and responsive. Additionally, custom control algorithms have been implemented to compensate for any temperature-induced variations in motor performance.

7. **TRIAS AND ANALYSIS**

7.1 Trial on Servo Motor Using RoboKits Software

To ensure that the servo motors could perform efficiently in the cold storage environment, initial trials were conducted using RoboKits software. This software simulation allowed us to test and evaluate the responsiveness, precision, and efficiency of the MG996R and SG90 servo motors under controlled conditions, simulating real-world tasks that the robotic arm will perform.

- **Process:**

The servo motors were interfaced with a microcontroller and controlled through the RoboKits software. Various movement tasks were simulated, including:

- Rotations: Testing the motor's ability to rotate at different speeds and angles.
- Gripping Actions: Simulating the process of gripping and releasing objects, crucial for handling tasks in cold storage.
- Lifting Tasks: Evaluating the torque capabilities of the motors in lifting items with varying weights.

The RoboKits software provided features for fine-tuning the motors' response to commands, including speed and precision. Additionally, temperature compensation settings were tested by simulating low-temperature conditions to assess how well the motors could adapt.

- **Analysis:**

During the trial, the servo motors responded accurately to the commands provided through RoboKits software. Key observations included:

- The MG996R motor, with its 11 kg/cm torque capacity, showed smooth movement and precision when lifting and rotating at various angles.
- The SG90 micro servo motor also performed as expected, handling lighter tasks with precision.
- Temperature compensation: The temperature simulations, ranging from -10°C to +60°C, showed that the motors maintained consistent performance across the expected cold storage temperature range. The servo motors did not exhibit any significant loss in precision or torque in these conditions.

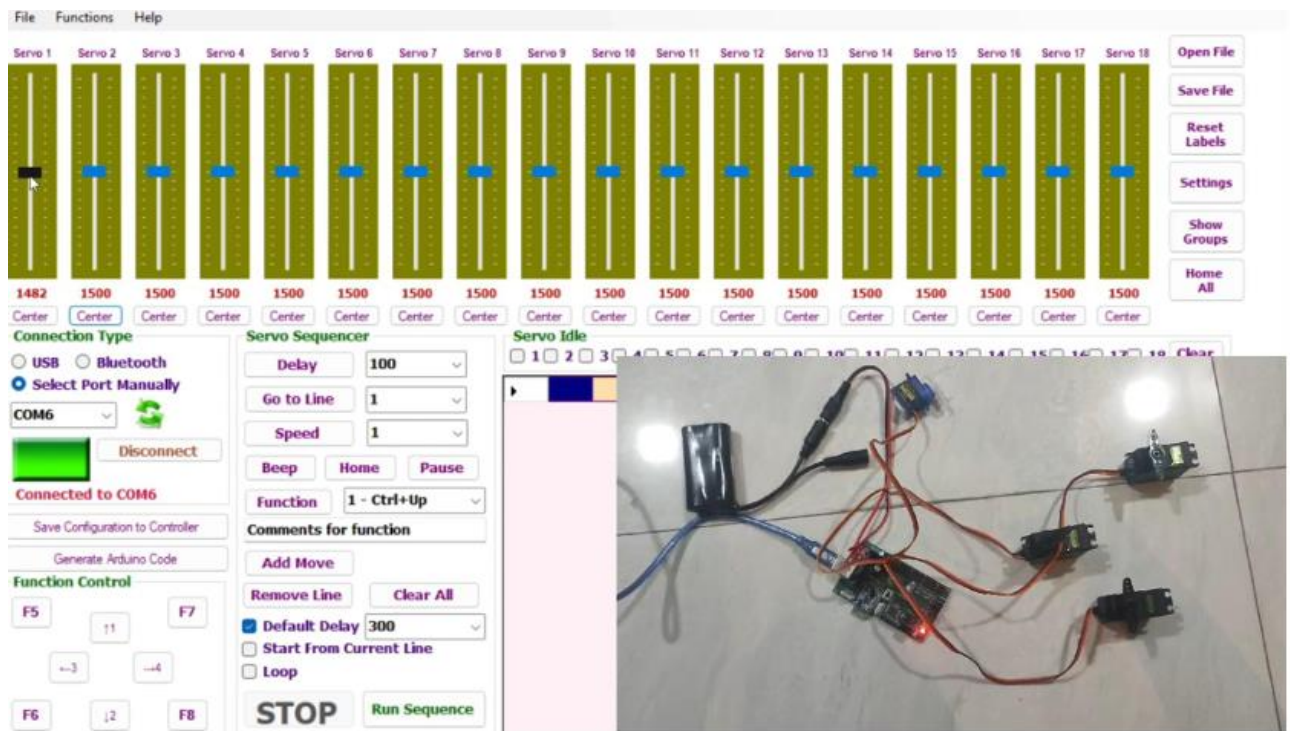


Fig.7.1. Trial on Servo Motor Using RoboKits Software

- Outcome:

The servo motors demonstrated excellent performance in terms of:

- Torque: Both motors exhibited the required torque for lifting and moving objects within the expected load limits.
- Movement Accuracy: The motors followed commands with high precision, meeting the project's requirements for robotic arm movements.
- Cold Storage Compatibility: Despite the temperature challenges, the servo motors continued to perform reliably within the specified temperature range of -10°C to +60°C, confirming their suitability for cold storage operations.

These results indicate that the servo motors selected for this project are capable of meeting the performance criteria in both normal and cold storage environments. This successful trial supports the next phase of development, where these motors will be integrated into the robotic arm for further testing in more realistic cold storage conditions.

8. CONCLUSION

The development of the Automated Mobile Robotic Arm (AMRA) for cold storage environments has addressed significant challenges associated with the handling and transportation of temperature-sensitive goods in extreme conditions. Through careful selection of components, including MG996R and SG90 servo motors, Samsung Li-Ion batteries, and advanced control systems, the project demonstrated that robotic automation could operate reliably in sub-zero temperatures.

Initial trials, including tests with RoboKits software, validated the system's ability to perform precise tasks such as gripping, lifting, and navigating in simulated cold storage conditions. The servo motors and control algorithms provided smooth, accurate, and responsive movements, ensuring the robotic arm meets the requirements of cold storage logistics. Additionally, material selection and power optimization strategies ensured durability, energy efficiency, and adaptability, even in harsh environments.

The AMRA offers a promising solution to enhance productivity, reduce manual labor, and improve worker safety by automating physically demanding tasks in cold storage facilities. By integrating mobility and high-precision robotics, the system can handle dynamic storage layouts while maintaining the care required for temperature-sensitive items.

Future iterations will focus on real-world deployment, incorporating advanced technologies like AI-based navigation, machine vision, and enhanced insulation techniques to further optimize the system for operational scalability and efficiency. This project sets a strong foundation for achieving fully automated and reliable cold storage solutions in the logistics and supply chain sectors.

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